

GCP Firewall Security Lab Report

Secure Remote Access Implementation using Google Cloud VPC Firewall Rules

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Role: Aspiring Cloud Security Engineer

Executive Summary

This project demonstrates implementation of secure network access controls in Google Cloud Platform (GCP) using VPC Firewall Rules and Compute Engine. The objective was to reduce attack surface by restricting SSH access (TCP/22) to a trusted public IP instead of allowing unrestricted Internet access.

Project Objectives

- Deploy a secure Compute Engine VM
- Restrict SSH access using firewall policies
- Apply network tags for segmentation
- Validate secure administrative access
- Document cloud security controls

Environment Details

Cloud Provider: Google Cloud Platform

Service: Compute Engine

OS: Debian GNU/Linux 12

VM: secure-vm

Machine Type: e2-micro

Region: asia-south1

Firewall Rule: allow-ssh-from-my-ip

Protocol: TCP/22

Security Risks Identified

Potential risks included brute-force attacks, unauthorized SSH access attempts, reconnaissance activity, and increased attack surface from publicly exposed administrative services.

Security Control Implementation

1. Enabled Compute Engine API
2. Created secure-vm instance
3. Applied network tag: ssh-secure
4. Identified trusted public IP
5. Created ingress firewall rule allowing TCP/22 only from trusted IP
6. Applied firewall rule to tagged VM
7. Validated SSH access using hostname, whoami and ip addr commands

Security Assessment

Principle of Least Privilege was implemented successfully. Access was limited to a single source IP, single protocol, single port, and a single VM. Network segmentation was enforced using tags and firewall rules.

Cloud Security Best Practices Applied

- Least Privilege Access
- Secure Administrative Access

- Network-Level Security Controls
- Attack Surface Reduction
- Resource Isolation
- Secure Resource Cleanup

Skills Demonstrated

Google Cloud Platform, Compute Engine Security, VPC Security, Firewall Management, SSH Security, TCP/IP Networking, Access Control, Security Validation, and Cloud Security Documentation.

Conclusion

The project successfully demonstrated secure remote administration in GCP through IP-restricted SSH access, VPC firewall controls, and network segmentation. The implementation aligns with industry cloud security best practices and showcases practical Cloud Security Engineer skills.